



IMSC

INTERNATIONAL MATHEMATICS SUMMER CAMP IMSC 2025
MOCK TEST 1-ALGEBRA

Date: Tuesday, 24th June 2024

Time: 13:30-18:00

Number of problems: 3

Total points: 21

PROBLEMS

Problem 1. Let a, b, c be positive reals such that $a^3 + b^3 + c^3 = a + b + c$. Show that

$$\frac{a^3}{a^4 + b + c} + \frac{b^3}{b^4 + c + a} + \frac{c^3}{c^4 + a + b} \leq 1.$$

Solution. By the Cauchy-Schwarz inequality we have

$$(a^4 + b + c) \left(\frac{1}{a^2} + b + c \right) \geq (a + b + c)^2.$$

This implies that

$$\frac{a^3}{a^4 + b + c} \leq \frac{1}{(a + b + c)^2} \cdot a^3 \left(\frac{1}{a^2} + b + c \right) = \frac{a + a^3(b + c)}{(a + b + c)^2}$$

and thus

$$\sum_{cyc} \frac{a^3}{a^4 + b + c} \leq \frac{1}{(\sum a)^2} \left(\sum a + (\sum a)(\sum a^3) - \sum a^4 \right) = 1 + \frac{\sum a - \sum a^4}{(\sum a)^2}.$$

Now, it is enough to show that $\sum a^4 \geq \sum a$. By the arithmetic-geometric mean inequality we have

$$2a^4 + a = a^4 + a^4 + a \geq 3\sqrt[3]{a^4 \cdot a^4 \cdot a} = 3a^3.$$

This implies that

$$\sum a^4 \geq \frac{3(\sum a^3) - \sum a}{2} = \sum a,$$

with equality if and only if $a = b = c = 1$. □

Marking scheme:

a. 3 points – for the application of Cauchy-Schwarz inequality

$$(a^4 + b + c) \left(\frac{1}{a^2} + b + c \right) \geq (a + b + c)^2$$

- b. 3 points – the proof of the inequality $\sum a^4 \geq \sum a$.
- c. 1 points – Finalizing the solution
- d. 7 points – complete solution stating the equality case as well

Problem 2. The Fibonacci sequence (F_n) is defined by $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. For any positive integer $n \geq 2$, prove that all coefficients of the polynomials

$$(1 - x^{F_2})(1 - x^{F_3})(1 - x^{F_4}) \cdots (1 - x^{F_n})$$

are in the set $\{-1, 0, 1\}$.

Solution. Denote $P_n(x) = (1-x)(1-x^2)(1-x^3)(1-x^5) \cdots (1-x^{F_n}) = a_0 + a_1x + \cdots + a_mx^m$, where $m = \deg P_n = F_{n+2} - 2$. Since $P_n(x) \equiv P_{n-1}(x) \pmod{x^{F_n}}$, using induction on n it suffices to prove that $|a_k| \leq 1$ for $k \geq F_n$. Moreover, $P_n(x) = x^m P_n(\frac{1}{x})$ implies that $a_k = (-1)^{n-1} a_{m-k}$, so it is enough to consider $\frac{1}{2}m \leq k < F_{n+1}$ only.

Since $P_n(x)(1+x^{F_{n-1}}+x^{F_n}+x^{2F_{n-1}}) = P_{n-2}(x)(1-x^{F_{n-1}})(1-x^{F_n})(1+x^{F_{n-1}}+x^{F_n}+x^{2F_{n-1}}) \equiv P_{n-2}(x) \pmod{x^{F_{n+1}}}$, and $\deg P_{n-2} < F_n$, equating coefficients yields

$$a_k = -a_{k-F_{n-1}} - a_{k-F_n} - a_{k-2F_{n-1}} \quad \text{for } k \geq F_n,$$

where we assume that $a_i = 0$ for $i < 0$. Thus, for $\frac{1}{2}m \leq k < F_{n+1}$ we have $F_{n-1} \leq k - F_{n-1} < F_n$, so the above relation gives us $a_{k-F_{n-1}} = -a_{k-F_n} - a_{k-2F_{n-1}} - a_{k-F_n-F_{n-2}}$. It follows that

$$a_k = (a_{k-F_n} + a_{k-2F_{n-1}} + a_{k-F_{n-1}-2F_{n-2}}) - a_{k-F_n} - a_{k-2F_{n-1}} = a_{k-F_n-F_{n-2}}.$$

By the inductive hypothesis, the latter is in the set $\{-1, 0, 1\}$, which completes the proof. $\square$

Marking scheme (additive):

- a. 2 points – using induction to reduce the problem to the inequality below
- b. 5 points – $|a_k| \leq 1$ for $k \geq F_n$.

Problem 3. A function $f: \mathbb{Z}^+ \rightarrow \mathbb{R}^+$ is non-decreasing and multiplicative: for any $a, b \in \mathbb{Z}^+$ such that $\gcd(a, b) = 1$, we have $f(ab) = f(a)f(b)$. Prove that there exists number $\alpha \geq 0$ such that $f(n) = n^\alpha$ for all positive integer n .

Solution. We first prove some lemmas:

Lemma 0.1. For positive integers a, b, c, d , $\frac{a}{b} < \frac{c}{d}$, we have $\frac{f(a)}{f(b)} \leq \frac{f(c)}{f(d)}$.

Proof of Lemma 1. Take positive integer p, q , both coprime to $abcd$ such that

$$\frac{a}{b} < \frac{p}{q} < \frac{c}{d},$$

then we have $bp > aq$ and $cq > dp$. With the multiplicative property, we then get

$$f(b)f(p) \geq f(a)f(q), \quad f(c)f(q) \geq f(d)f(p),$$

from which Lemma 1 follows.

Lemma 0.2. Given positive integer M , let S be the set of positive integers coprime to M , then the set

$$T = \left\{ \frac{f(a)}{f(b)} \mid a, b \in S \right\}$$

is dense in $\mathbb{R}^+$.

Proof of Lemma 2. Given any $\delta > 0$, take positive integer k such that $(1 + \delta)^k \leq 2 < (1 + \delta)^{k+1}$. Then choose pairwise coprime odd integers $a_1, \dots, a_k, b_1, \dots, b_k$ such that $\frac{a_i}{b_i} \in (1 + \delta, 1 + 2\delta)$. We have

$$\prod_{i=1}^k \frac{a_i}{b_i} < (1 + 2\delta)^k < (1 + \delta)^{2k} \leq 4$$

and then

$$\prod_{i=1}^k \frac{f(a_i)}{f(b_i)} = \frac{f(a_1 a_2 \cdots a_k)}{f(b_1 b_2 \cdots b_k)} \leq \frac{f(4b_1 b_2 \cdots b_k)}{f(b_1 b_2 \cdots b_k)} = f(4).$$

With Lemma 1, this implies

$$\min \frac{f(a_i)}{f(b_i)} \leq f(4)^{1/k}. \quad (1)$$

The set S contains a subset $\{Mt + 1\}$ whose consecutive terms has ratio $\frac{M(t+1)+1}{Mt+1} < 1 + \delta$ for $t \gg 0$ and then use Lemma 1 and (1) to get $\frac{f(M(t+1)+1)}{f(Mt+1)} \in [1, f(4)^{1/k}]$. Take $\delta \rightarrow 0^+$, $k \rightarrow \infty$, we have T is dense. Since ratio of terms in S is unbounded, T is dense in $\mathbb{R}^+$.

Lemma 0.3. If $\frac{a}{b} = \frac{c}{d}$, then $\frac{f(a)}{f(b)} = \frac{f(c)}{f(d)}$.

Proof of Lemma 3. Assume the contrary, if $\frac{f(a)}{f(b)} < \frac{f(c)}{f(d)}$, use Lemma 2 to get pq coprime to $abcd$ such that

$$\frac{f(a)}{f(b)} < \frac{f(p)}{f(q)} < \frac{f(c)}{f(d)}.$$

Use the multiplicative property to get

$$f(aq) < f(bp), \quad f(dp) < f(cq).$$

Use the non-decreasing property to get

$$aq < bp, \quad dp < cq \Rightarrow \frac{a}{b} < \frac{p}{q} < \frac{c}{d}$$

contradiction.

Now back to proof of the problem. For any prime number p and positive integer k , with Lemma 3 we have

$$\frac{f(p^k)}{f(p^{k-1})} = \frac{f(p)}{f(1)},$$

by induction to get $f(p^k) = f(p)^k$. For two prime number p, q and positive integer k, l such that $p^k < q^l$, we have

$$f(p^k) \leq f(q^l) \Rightarrow f(p)^k \leq f(q)^l.$$

That is

$$\frac{k}{l} < \frac{\ln q}{\ln p} \Rightarrow \frac{k}{l} \leq \frac{\ln f(q)}{\ln f(p)}$$

holds for all rational number $\frac{k}{l}$, switching p, q to get the $\geq$ version of inequality, this then implies $\frac{\ln q}{\ln p} = \frac{\ln f(q)}{\ln f(p)}$ for any prime number p, q , then $\frac{\ln f(p)}{\ln p}$ is a constant, say α , for prime p . Thus $f(p) = p^\alpha$ and then $f(n) = n^\alpha$ for all $n \in \mathbb{Z}^+$.

Marking scheme(additive):

• 5 points -

- [a] 2 points - Lemma 1
- [b] 2 point - Lemma 2
- [c] 3 points - Lemma 3